

6.5 - Equation of a Line Given Slope and a Point

DO IT NOW!

Instructions: Determine the equation of the line, in slope y-intercept form, that has a slope of 3 and goes through the point (2, -5)

Note: You can write the equation of a line once you know the slope and y-intercept.

$$y = mx + b$$

Slope y-intercept

Step 1: State what you know about the line

$$\text{Slope} = m = 3$$

$$\text{Point on line: } (x, y) = (2, -5)$$

Step 2: Determine the y-intercept of the line

To do this we can use the equation $y = mx + b$, substitute in values for m , x and y and then solve for the b value. Use the point on the line that is given for the x and y values.

$$\begin{aligned} y &= mx + b \\ -5 &= 3(2) + b \\ -5 &= 6 + b \\ -5 - 6 &= b \\ b &= -11 \end{aligned}$$

STEP 3: Write the equation of the line in slope y-intercept form.

$$y = 3x - 11$$

Note: When writing the final equation of the line, plug in values for m and b , not for x and y .

Example 1: Find the equation of the line with a slope of $\frac{1}{2}$ that passes through (1, 5).

Step 1: State what you know about the line

$$\text{slope} = m = \frac{1}{2}$$

$$\text{Point: } (\overset{x}{1}, \overset{y}{5})$$

Step 2: Determine the y-intercept of the line

$$y = mx + b$$

$$5 = \frac{1}{2}(1) + b$$

$$5 = \frac{1}{2} + b$$

$$\frac{5}{1} - \frac{1}{2} = b$$

$$\frac{10}{2} - \frac{1}{2} = b$$

$$b = \frac{9}{2}$$

STEP 3: Write the equation of the line in slope y-intercept form.

$$y = \frac{1}{2}x + \frac{9}{2}$$

Example 2: Find the equation of the line with a slope of 3 and that passes through (0, 2). Then graph the line.

Step 1: State what you know about the line

$$\text{slope: } m = 3$$

$$\text{Point: } (\overset{x}{0}, \overset{y}{2})$$

↑
This point is the y-intercept.
If you notice this, you don't
have to calculate it.

Step 2: Determine the y-intercept of the line

$$y = mx + b$$

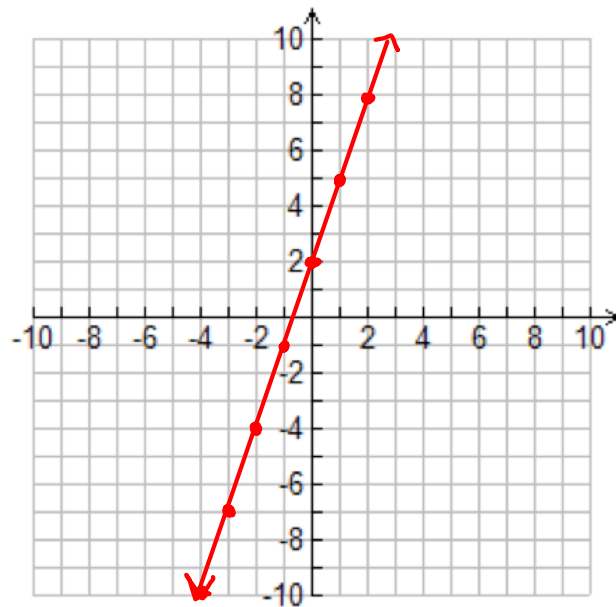
$$2 = 3(0) + b$$

$$2 = b$$

STEP 3: Write the equation of the line in slope y-intercept form.

$$y = 3x + 2$$

Step 4: Graph the line using the slope and y-intercept



Example 3: Determine the equation of a line that is parallel to the line $y = -2x - 7$ and passes through the point $(1, -3)$.

Step 1: State what you know about the line

Remember: lines that are parallel have the same slope. They do not have the same y-intercept. You will still have to solve for that.

slope: $m = -2$

Point: $(\overset{x}{1}, \overset{y}{-3})$

Step 2: Determine the y-intercept of the line

$$y = mx + b$$

$$-3 = -2(1) + b$$

$$-3 = -2 + b$$

$$-3 + 2 = b$$

$$-1 = b$$

STEP 3: Write the equation of the line in slope y-intercept form.

$$y = -2x - 1$$

Example 4: Determine the equation of a line that is perpendicular to the line $2x - y + 4 = 0$ and passes through the point $(-2, 5)$.

Hint: to determine the slope you will need to put the equation into $y = mx + b$ form so that you can see the slope and then take the negative reciprocal.

$$2x - y + 4 = 0$$

$$2x + 4 = y$$

$$y = 2x + 4$$

STEP 1: state what you know about the line

Slope of given line: $m = 2$

Slope of perpendicular line (find negative reciprocal):

$$\underline{b} m = -\frac{1}{2}$$

Point on the perpendicular line: $(\overset{x}{-2}, \overset{y}{5})$

Step 2: Determine the y-intercept of the line (make sure to use the slope of the perpendicular line)

$$y = mx + b$$

$$5 = -\frac{1}{2}(-2) + b$$

$$5 = \frac{2}{2} + b$$

$$5 = 1 + b$$

$$5 - 1 = b$$

$$4 = b$$

STEP 3: Write the equation of the line in slope y-intercept form (make sure to use the slope of the perpendicular line).

$$y = -\frac{1}{2}x + 4$$

Consolidation:

To write the equation of a line you need to know the slope (m) and y-intercept (b).

You can use the slope of a line and a point on the line to calculate the y-intercept.

To find the slope of a perpendicular line, find the negative reciprocal.